Tenemos :

$$\left.\begin{array}{l} \bullet\ \underset{n^{[l]}\times m}{Z^{[l]}} = \underset{n^{[l]}\times n^{[l-1]}}{\Theta^{[l]}}\ \underset{n^{[l-1]}\times m}{A^{[l-1]}} + \underset{n^{[l]}\times m}{b^{[l]}} \\ \\ \bullet\ \underset{n^{[l]}\times m}{A^{[l]}} = \sigma\left(Z^{[l]}\right) \end{array}\right\} \text{layer } l$$

$$\left.\begin{array}{l} \bullet\ \underset{1\times m}{Z^{[L]}} = \underset{1\times n^{L-1}}{\Theta^{[L]}}\ \underset{n^{L-1}\times m}{A^{[L-1]}} + \underset{1\times m}{b^{[L]}} \\ \\ \bullet\ \underset{1\times m}{A^{[L]}} = \sigma\left(Z^{[L]}\right) \end{array}\right\} \text{layer } L$$

Además, la fun. coste $\quad J = \frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log A^L + (1-y^{(i)})\log(1-A^L)\right]$

→ Cálculo $\frac{\partial J}{\partial \Theta}$

$$\frac{\partial J}{\partial \Theta^L} = \frac{\partial J}{\partial A^L}\,\frac{\partial A^L}{\partial z^L}\,\frac{\partial z^L}{\partial \Theta}$$

Recordar: $\sigma = \frac{1}{1+e^{-z}} = (1+e^{-z})^{-1}$

$$\sigma' = (1+e^{-z})^{-2}\, e^{-z}$$

$$\sigma' = \frac{e^{-z}}{(1+e^{-z})(1+e^{-z})}$$

$$\sigma' = \sigma\left(\frac{e^{-z}}{1+e^{-z}}\right)$$

$$\sigma' = \sigma\left(1 - \frac{1}{1+e^{-z}}\right)$$

como $A^{[L]} = \frac{1}{1+e^{-z^{[L]}}}$

$$1 - A^{[L]} = 1 - \frac{1}{1+e^{-z^{[L]}}}$$

$$\Rightarrow \frac{\partial J}{\partial \Theta^L} = \frac{1}{m}\sum_{i=1}\left[\frac{y^{(i)}}{A^L} - \frac{(1-y^{(i)})}{1-A^L}\right] A^L(1-A^L)\ \frac{\partial z^L}{\partial \Theta^L}$$

↪ $z^l = \Theta A^{l-1} + b^l$

$$J = \frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log A^L + (1-y^{(i)})\log(1-A^L)\right]$$

$$\frac{\partial J}{\partial \Theta^l} = \frac{1}{m}\sum_{i=1}\left[\frac{y^{(i)}}{A^l} - \frac{(1-y^{(i)})}{1-A^l}\right] A^l(1-A^l) \cdot A^{l-1\,T}$$

$$= \frac{1}{m}\sum_{i=1}\left[\frac{y^{(i)}(1-A^l) - (1-y^{(i)})A^l}{\cancel{A^l}\;\cancel{(1-A^l)}}\right]\cancel{A^l}\;\cancel{(1-A^l)}\;A^{l-1\,T}$$

$$= \frac{1}{m}\sum_{i=1}\left(y^{(i)}(1-A^l) - (1-y^{(i)})A^l\right)A^{l-1\,T}$$

$$= \frac{1}{m}\sum_{i=1}\left(y^{(i)} - \cancel{y^{(i)}A^l} - A^l + \cancel{y^{(i)}A^l}\right)A^{l-1\,T}$$

$$\underset{1\times n^{l-1}}{\frac{\partial J}{\partial \Theta^l}} = \frac{1}{m}\sum_{i=1}\left(\underset{1\times m}{y^{(i)}} - \underset{1\times m}{A^l}\right)\underset{m\times n^{l-1}}{A^{l-1\,T}}$$

→ $$\frac{\partial J}{\partial b^l} = \frac{\partial J}{\partial A^l}\,\frac{\partial A^l}{\partial z^l}\,\cancel{\frac{\partial z^l}{\partial b^l}} \rightarrow 1$$

$$= \frac{1}{m}\sum_{i=1}\left[\frac{y^{(i)}}{A^l} - \left(\frac{1-y^{(i)}}{1-A^l}\right)\right]A^l(1-A^l)$$

$$\underset{1\times m}{\frac{\partial J}{\partial b^l}} = \frac{1}{m}\sum_{i=1}\left(\underset{1\times m}{y^{(i)}} - \underset{1\times m}{A^l}\right)$$

→ $$\frac{\partial J}{\Theta^{l-1}}$$

Tener en cuenta $$J = \frac{1}{m}\sum^{m}\left[y^{(i)}\log A^l + (1-y^{(i)})\log(1-A^l)\right]$$

$$A^{l-1} = \sigma(z^{l-1})$$

$$z^{l-1} = \Theta^{l-1}A^{l-2} + b^{l-1}$$

$$\frac{\partial J}{\partial \Theta^{l-1}} = \frac{\partial J}{\partial A^{l-1}}\,\frac{\partial A^{l-1}}{\partial z^{l-1}}\,\frac{\partial z^{l-1}}{\partial \Theta^{l-1}}$$

- $$\frac{\partial J}{\partial A^{l-1}} = \frac{\partial J}{\partial z^l}\,\frac{\partial z^l}{\partial A^{l-1}}$$

$$= \underbrace{\frac{\partial J}{\partial A^L} \frac{\partial A^L}{\partial Z^L}}_{\text{ya lo había calculado}} \cdot \frac{\partial Z^L}{\partial A^{L-1}}$$

$$= \frac{1}{m} \sum \left[\frac{y^{(i)}}{A^L} - \frac{(1-y^{(i)})}{1-A^L} \right] A^L (1-A^L) \quad \frac{\partial Z^L}{\partial A^{L-1}}$$

$$= \frac{1}{m} \sum \left\{ y^{(i)} - \cancel{y^{(i)} A^L} - A^L + \cancel{A^L y^{(i)}} \right\} \frac{\partial Z^L}{\partial A^{L-1}}$$

$$= \frac{1}{m} \sum (y^{(i)} - A^L) \underbrace{\frac{\partial Z^L}{\partial A^{L-1}}}_{\text{como } Z^L = \Theta^L A^{L-1} + b^L}$$

$$\frac{\partial J}{\partial A^{L-1}} = \frac{1}{m} \sum (y^{(i)} - A^L) \, \Theta^{L^T}$$

- $\frac{\partial A^{L-1}}{\partial Z^{L-1}} = \sigma'(Z^{L-1})$

  $\frac{\partial A^{L-1}}{\partial Z^{L-1}} = A^{L-1}(1 - A^{L-1})$

- $\frac{\partial Z^{L-1}}{\partial \Theta^{L-1}} = A^{L-2^T}$

$$\Rightarrow \frac{\partial J}{\partial \Theta^{L-1}} = \frac{1}{m} \overbrace{\sum (y^{(i)} - A^L) \, \Theta^{L^T}}^{\partial J / \partial A^{L-1}} A^{L-1} (1 - A^{L-1}) \, A^{L-2^T}$$

$$\text{llamo } \delta^{L-1} = \frac{\partial J}{\partial A^{L-1}} A^{L-1} (1 - A^{L-1})$$

$$\frac{\partial J}{\partial \Theta^{L-1}} = \frac{1}{m} \sum \delta^{L-1} \cdot A^{L-2^T}$$

$$\rightarrow \frac{\partial J}{\partial b^{L-1}} = \frac{1}{m}\sum_{i=1}^{m} \delta^{L-1}$$

$$\bullet \quad \frac{\partial J}{\partial \Theta^{\ell}} = \frac{\partial J}{\partial A^{\ell}} \frac{\partial A^{\ell}}{\partial z^{\ell}} \frac{\partial z^{\ell}}{\partial \Theta^{\ell}}$$

donde $A^{\ell} = \sigma(z^{\ell})$

$z^{\ell} = \Theta^{\ell} A^{\ell-1} + b^{\ell} \rightarrow z^{\ell+1} = \Theta^{\ell+1} A^{\ell} + b^{\ell}$

$J = \frac{1}{m} \sum^{m} \left[y^{(i)} \log A^{L} + (1-y^{(i)}) \log(1-A^{L}) \right]$

$$\rightarrow \frac{\partial J}{\partial A^{\ell}} = \frac{\partial J}{\partial z^{\ell+1}} \frac{\partial z^{\ell+1}}{\partial A^{\ell}}$$

$$= \frac{\partial J}{\partial z^{\ell+1}} \Theta^{\ell+1\,T}$$

$$\rightarrow \frac{\partial A^{\ell}}{\partial z^{\ell}} = A^{\ell}(1-z^{\ell})$$

$$\rightarrow \frac{\partial z^{\ell}}{\partial \Theta^{\ell}} = A^{\ell-1\,T}$$

$$\Rightarrow \frac{\partial J}{\partial \Theta^{\ell}} = \frac{\partial J}{\partial z^{\ell+1}} \Theta^{\ell+1\,T} A^{\ell}(1-z^{\ell}) A^{\ell-1\,T}$$

$$\frac{\partial J}{\partial \Theta^{\ell}} = \frac{1}{m} \delta^{\ell+1} \Theta^{\ell+1\,T} A^{\ell}(1-z^{\ell}) A^{\ell-1\,T}$$

definimos $\delta^{\ell} = \Theta^{\ell+1\,T} \delta^{\ell+1} \cdot A^{\ell}(1-z^{\ell})$

$$\frac{\partial J}{\partial W^{l}} = \frac{1}{m} \delta^{l} A^{l-1^{T}}$$

- $$\frac{\partial J}{\partial b^{l}} = \frac{1}{2} \delta^{l}$$